UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Flowcharts and visuals

Procedures, decisions, concept maps and comparison tables.

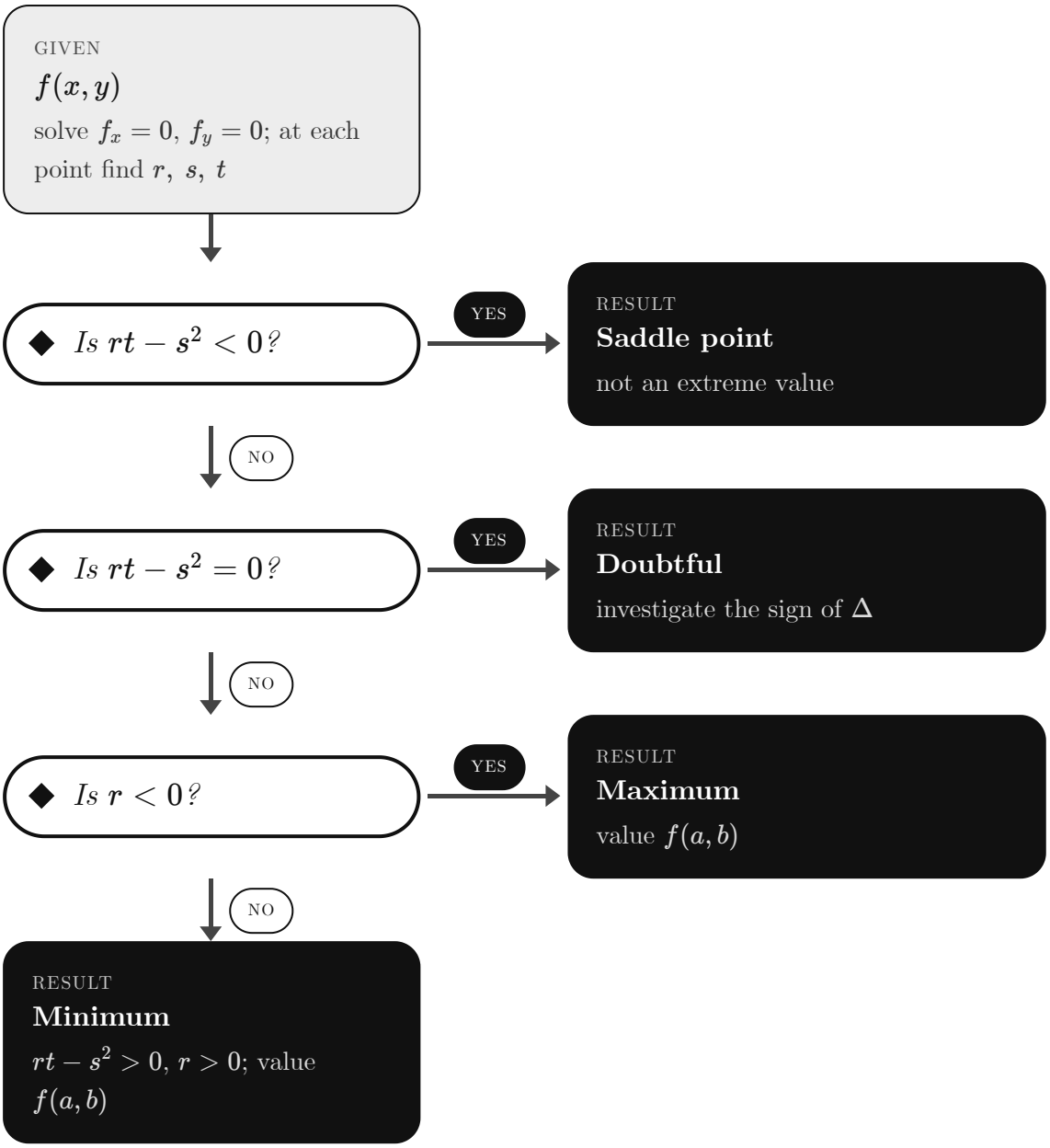
Class 009 · Introduction to maxima and minima for a function of two variables · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S009·7	Deck 009, slide 7 (rendered .pptx)
T009 00:07:04	AV-summary transcript 009 (“Extremas of functions”), segment start
CW U1 P31	Unit 1 class-work problem 31
ESA Dec-23 1c	Past paper: Dec 2023, question 1.c

Contents

1. Figure F3.1 · Working rule: nature of a stationary point
2. Figure F3.2 · From one variable to two
3. The four outcomes of the test

3 Conditions and the working rule



Remark. Repeat the ladder for every stationary point.

Exam trap. Never classify from r alone.

Figure F3.1. Working rule: nature of a stationary point ESA ★★★

S009·8 · S009·9 · T009 00:11:32

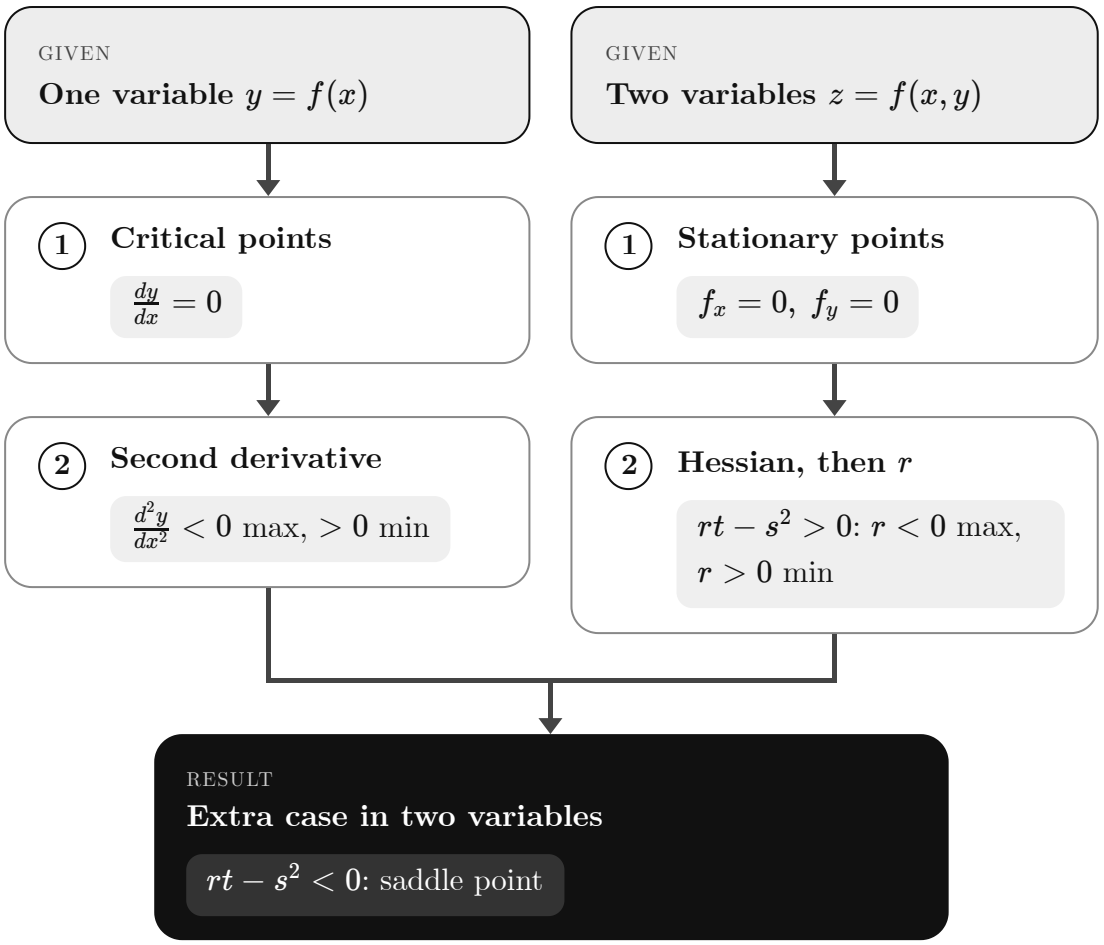


Figure F3.2. From one variable to two ESA ★★

T009 00:06:00 · T009 00:10:20

The four outcomes of the test

$rt - s^2$	r	Nature of (a, b)	Example at the origin
> 0	< 0	Maximum	$-x^2 - y^2$
> 0	> 0	Minimum	$x^2 + y^2$ (W3.1)
< 0	any	Saddle point	$x^2 - y^2$ (W3.2)
$= 0$	any	Doubtful: further investigation	$x^4 + y^4$ (min), $x^4 - y^4$ (neither)

S009·7 · S009·8 · S009·9